

Small intestine

Department of Anatomy

The abdomen (5)

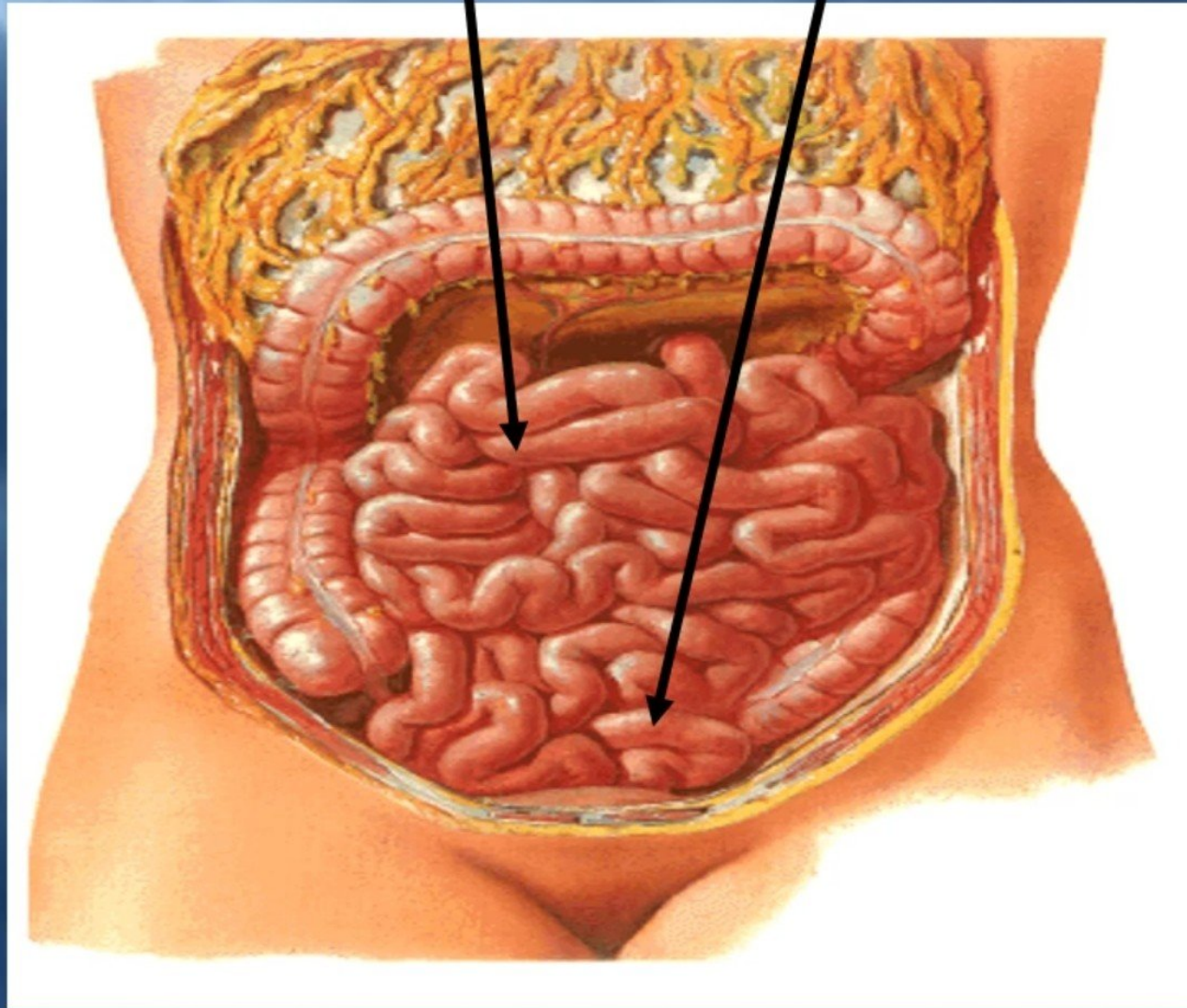
Third Part

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Lecture Points

- **Structure of intestine**
- **Five main functions of the bowel (intestine).**
- **Parts of the Small Intestine.**
- **The Duodenum (Location and Description)**
 - ✓ **Parts.**
 - ✓ **Length.**
 - ✓ **Direction.**
 - ✓ **Relation to Peritoneum.**
 - ✓ **First part of Duodenum.**
 - ✓ **Second part of Duodenum**
 - ✓ **Third part of Duodenum**
 - ✓ **Fourth part of Duodenum.**
 - ✓ **Nerve Supply of Duodenum**
 - ✓ **Arterial Supply of Duodenum**
 - ✓ **Veins drainage of Duodenum.**
 - ✓ **Lymph drainage of duodenum.**
- **The Jejunum & Ileum.**
- **Jejunum and Ileum (Location and Description).**
- **The root of mesentery of the small intestine**
- **The duodenojejunal flexure.**
- **The ileocecal junction.**
- **The anatomical features of Jejunum & Ileum**
- **Difference Between Jejunum & Ileum.**
- **Blood Supply of Jejunum & Ileum.**
- **Nerve Supply of Jejunum & Ileum**
- **Lymph drainage of Jejunum & Ileum**

Jejunum & Ileum

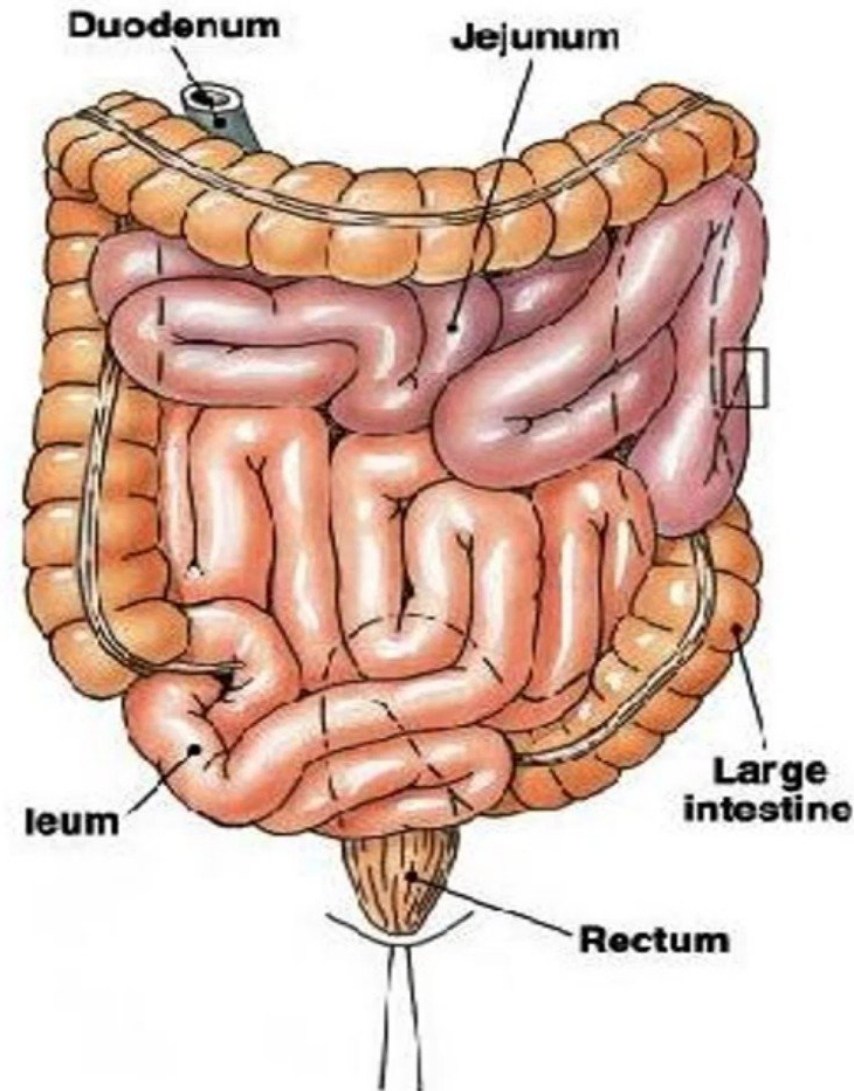


Jejunum and Ileum

Location and Description

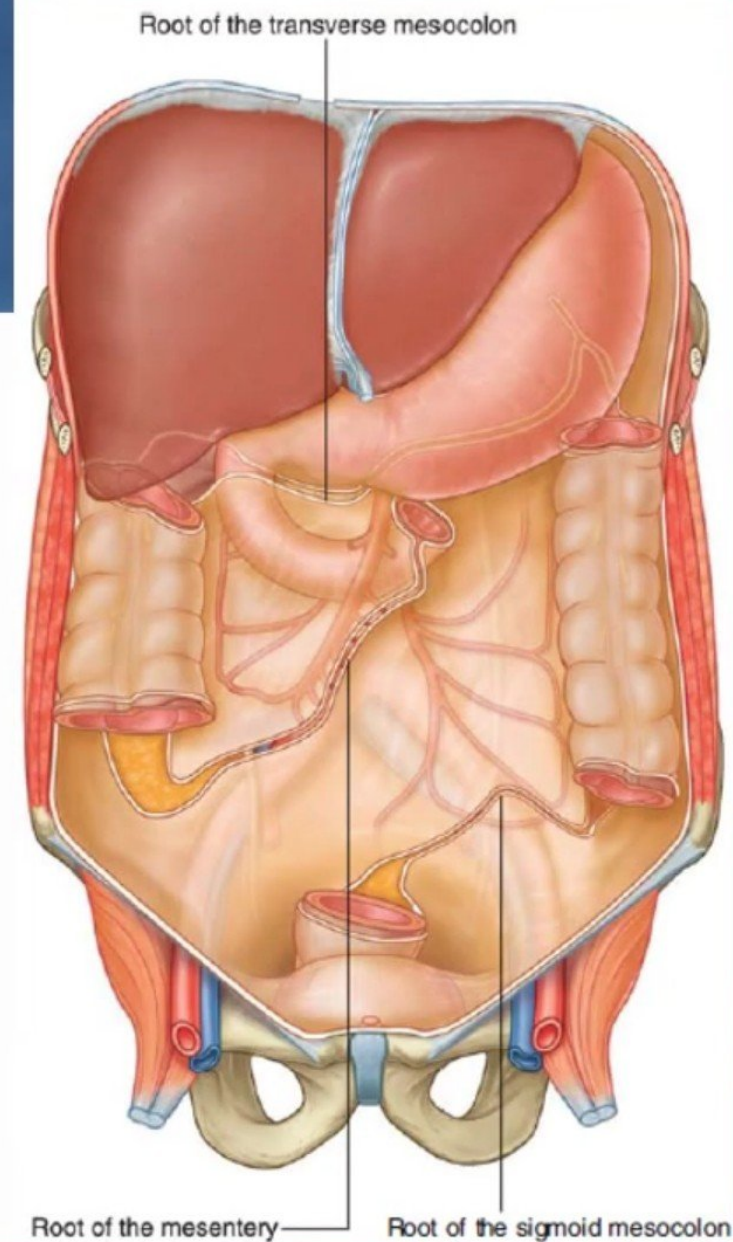
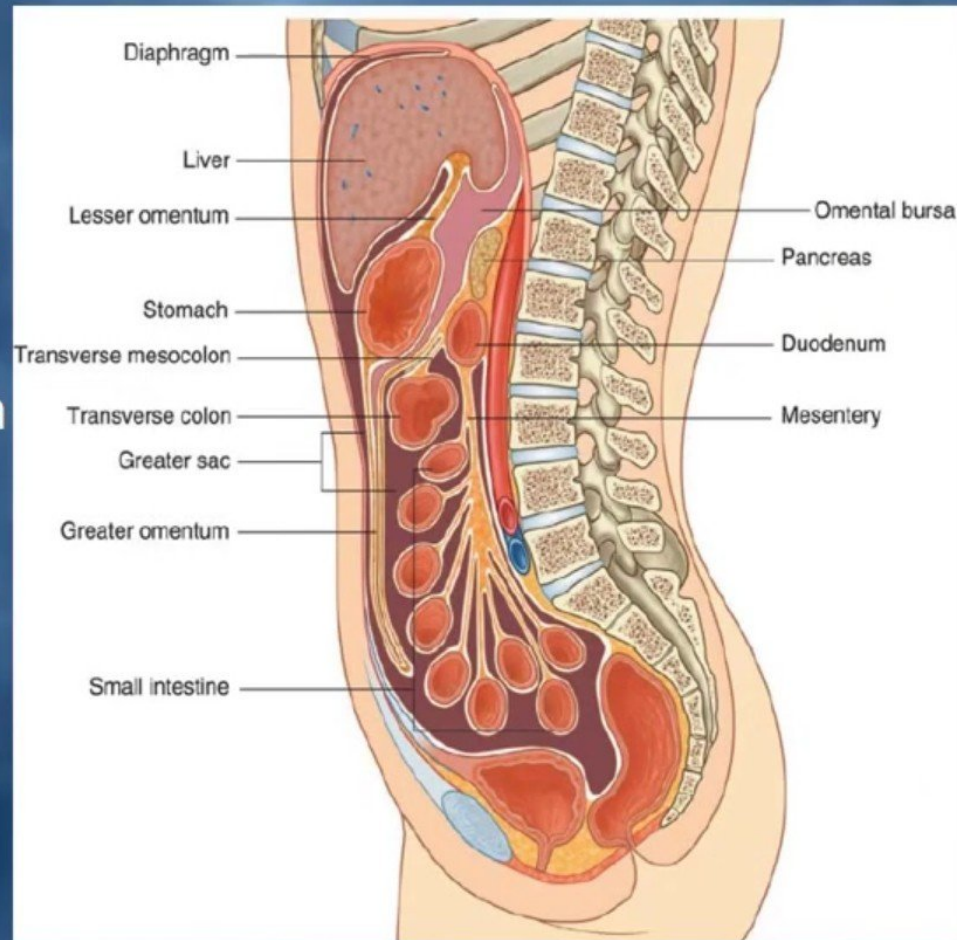
- The jejunum and ileum measure about **20 ft (6 m) long**.
- The **upper two fifths** is the **jejunum** & the **lower 3/5** is the **ileum**.
- The jejunum begins at **the duodenojejunal flexure**.
- the ileum ends at **the ileocecal junction**.

Regions of the Small Intestine



The root of mesentery of the small intestine

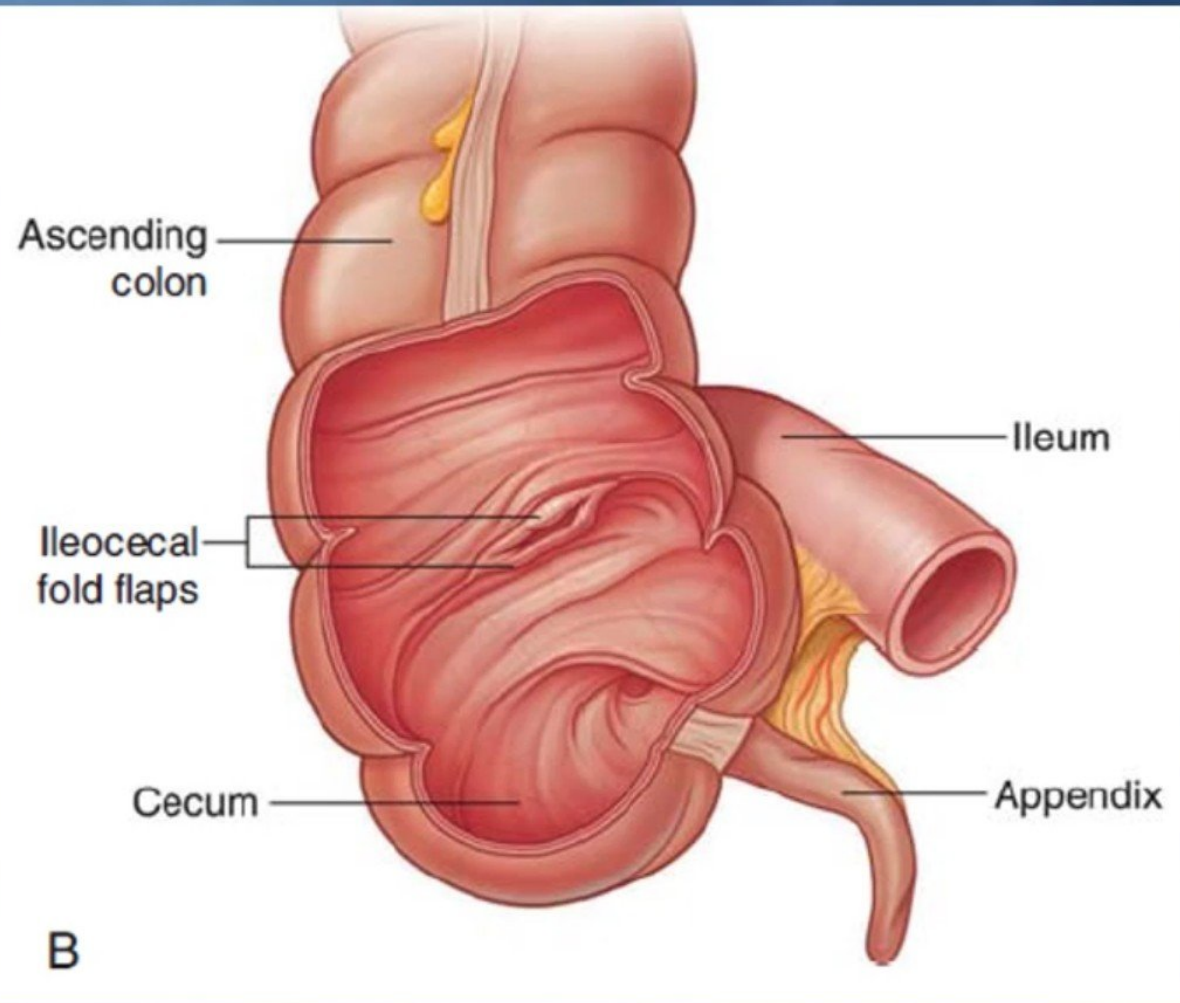
The **coils of jejunum and ileum** are **freely mobile** and are attached to the posterior abdominal wall by a fan-shaped double layered fold of peritoneum known as **the root of mesentery of the small intestine**.



The duodenojejunal flexure

It is surrounded by a fold of peritoneum containing muscle fibers called the **suspensory muscle (ligament)** of duodenum (ligament of Treitz).

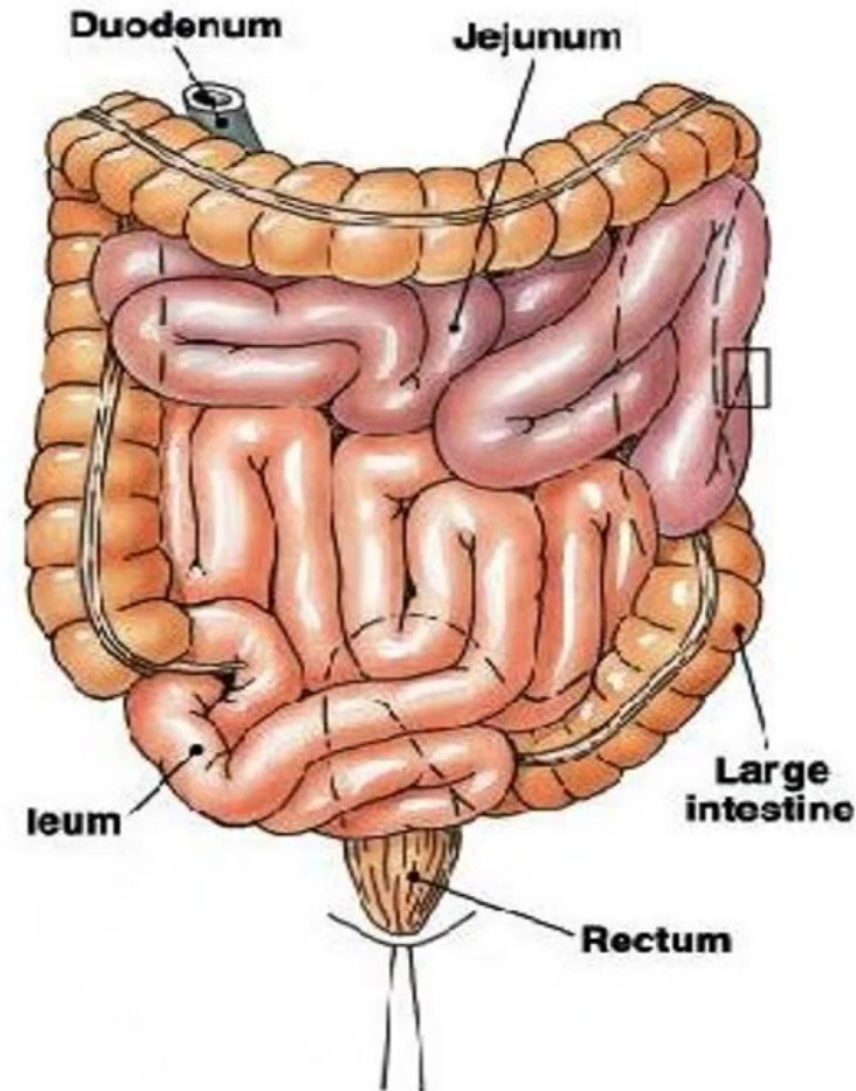
The ileocecal junction



The anatomical features of Jejunum

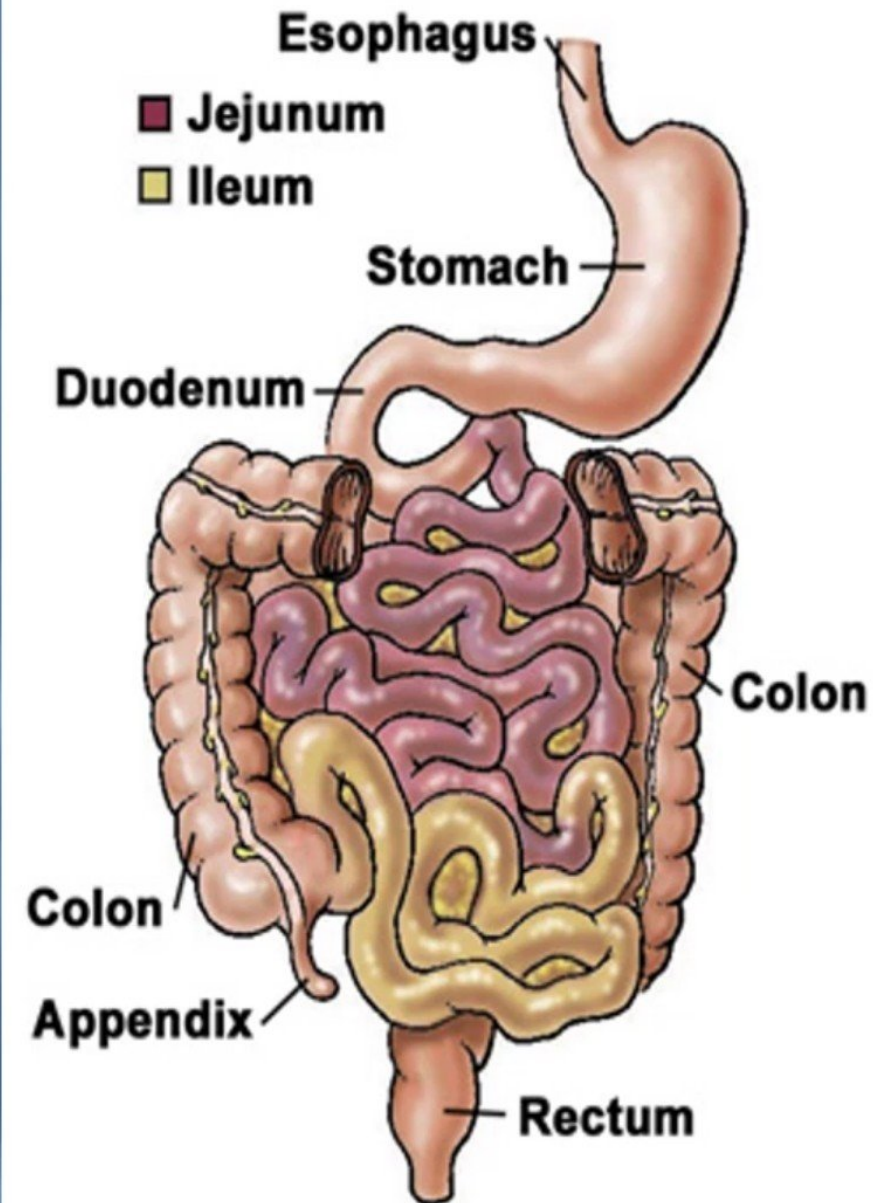
- The jejunum is the **main section of the small intestine**.
- It represents the **proximal two fifths of small intestine**.
- It covers about 15 feet and is responsible for the absorption of almost all nutrients except water.

Regions of the Small Intestine



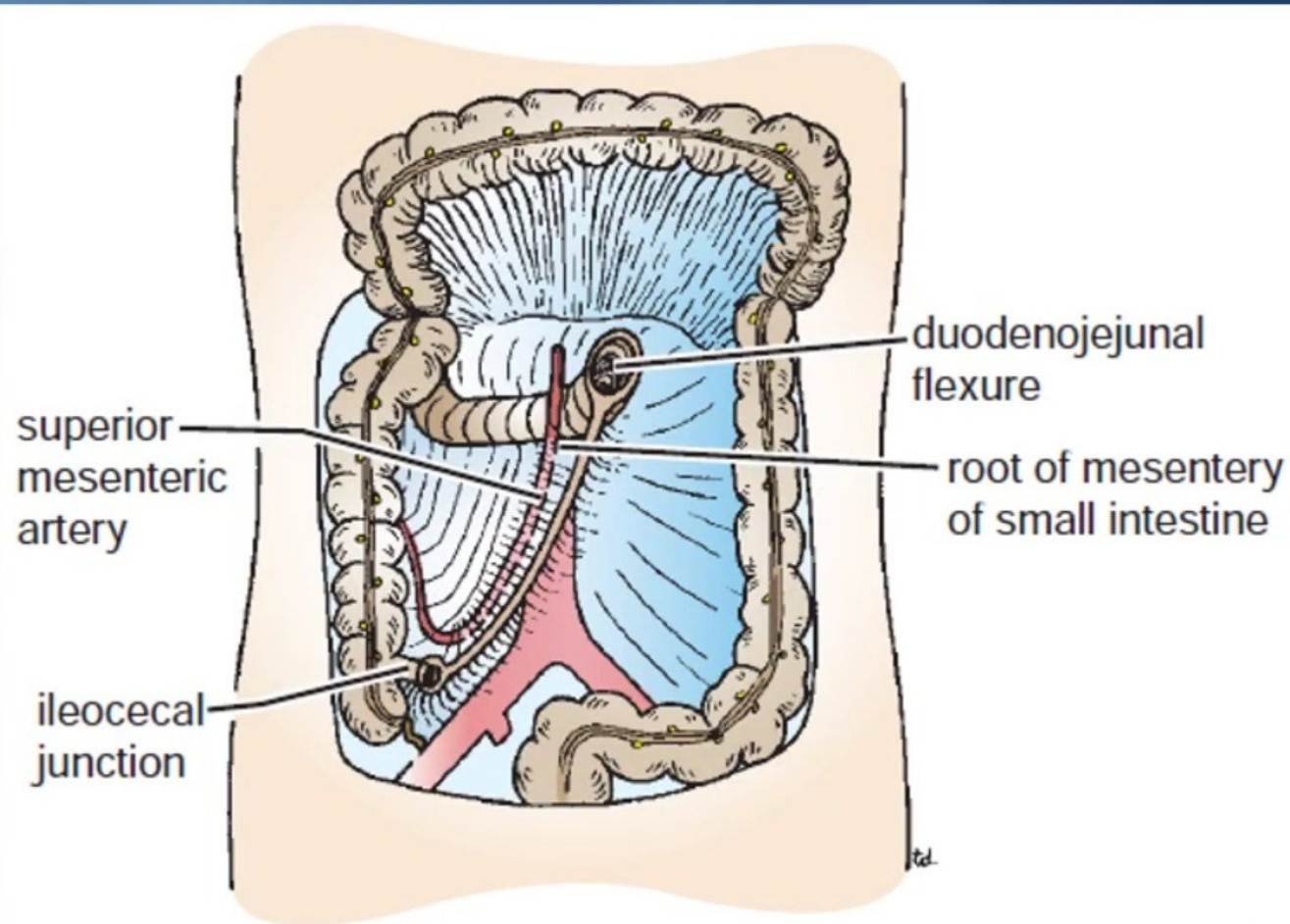
The anatomical features of ileum

- The ileum is the **last section of the small intestine** and spans about 6 feet.
- Its function is to **absorb water and vitamins**.



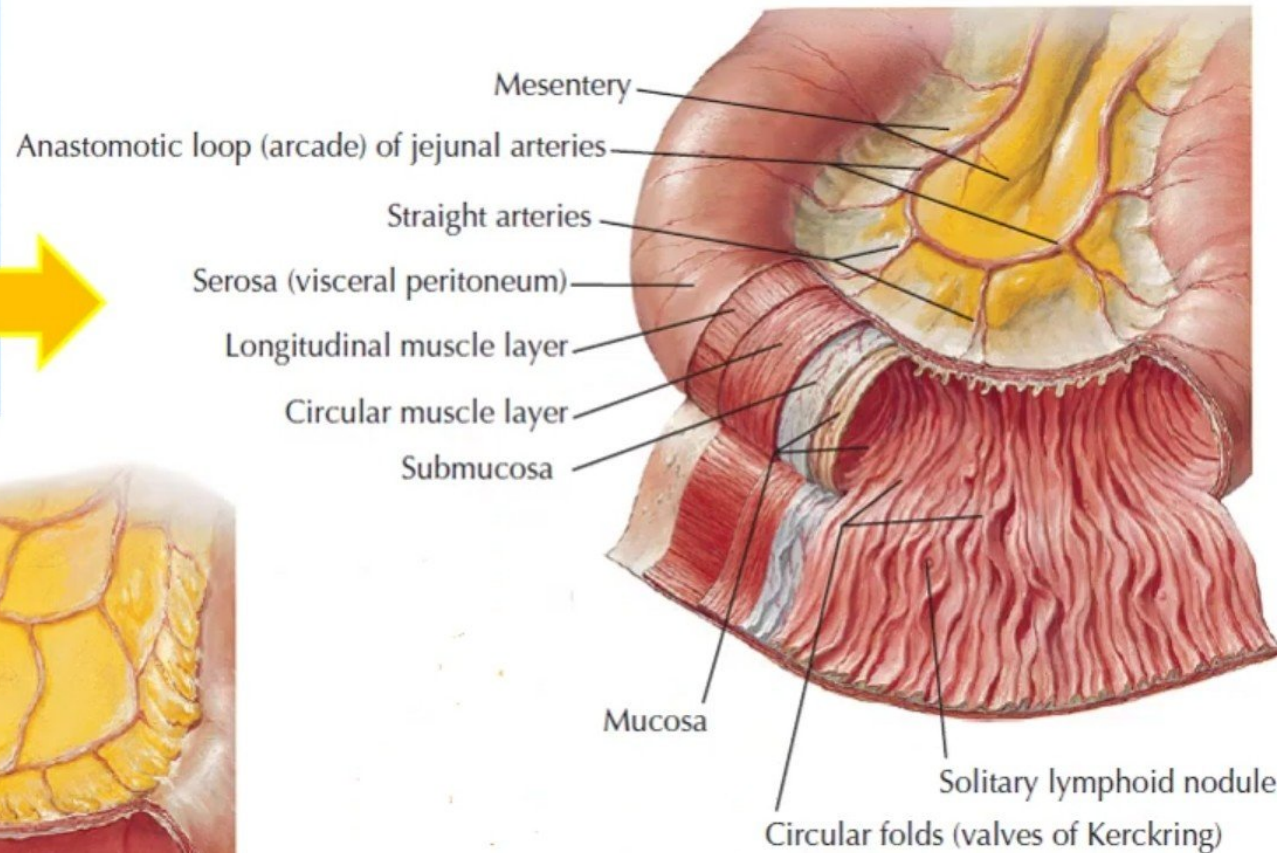
The mesentery

- The coils of **jejunum and ileum** are freely mobile and attachment of **the root of the mesentery** of the small intestine to the posterior abdominal wall.
- Note that it extends from the **duodenojejunal flexure** on left of the aorta, downward, and to the right to **the ileocecal junction**.
- **The superior mesenteric artery** lies in **the root of the mesentery**.

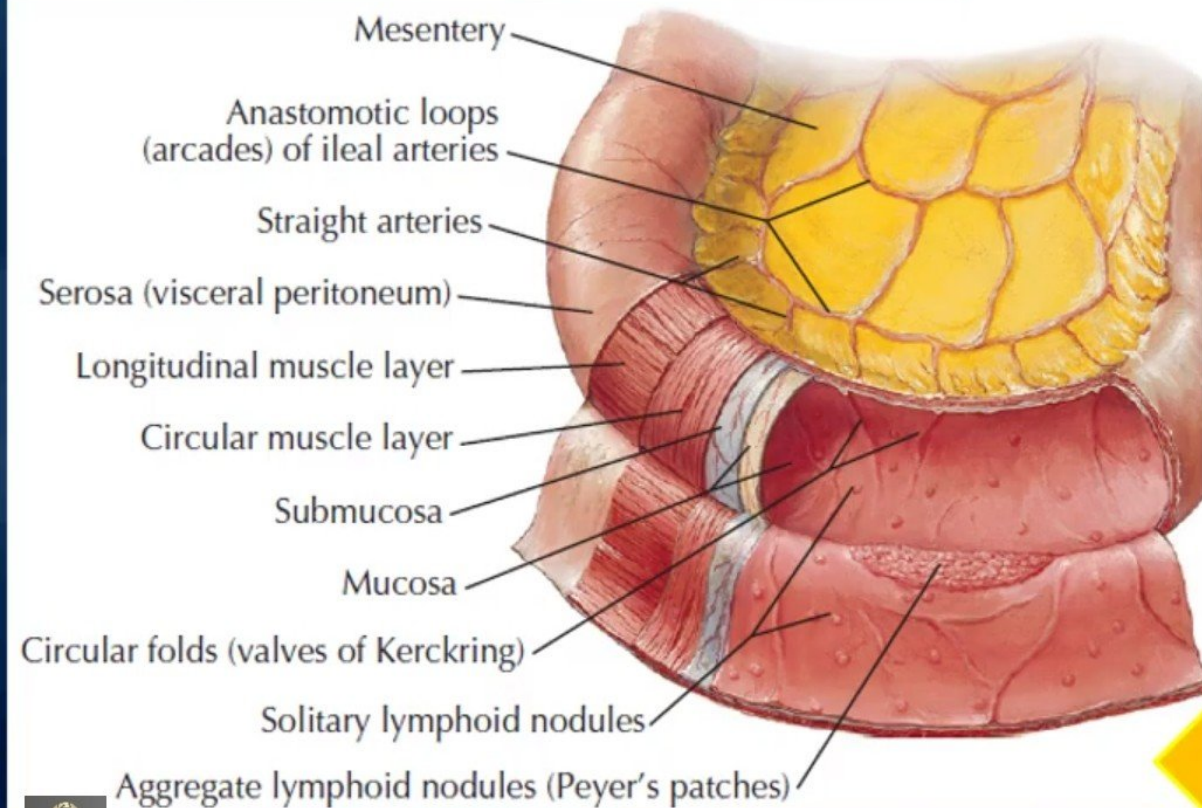


The anatomical features of Jejunum & Ileum

Jejunum



Ileum

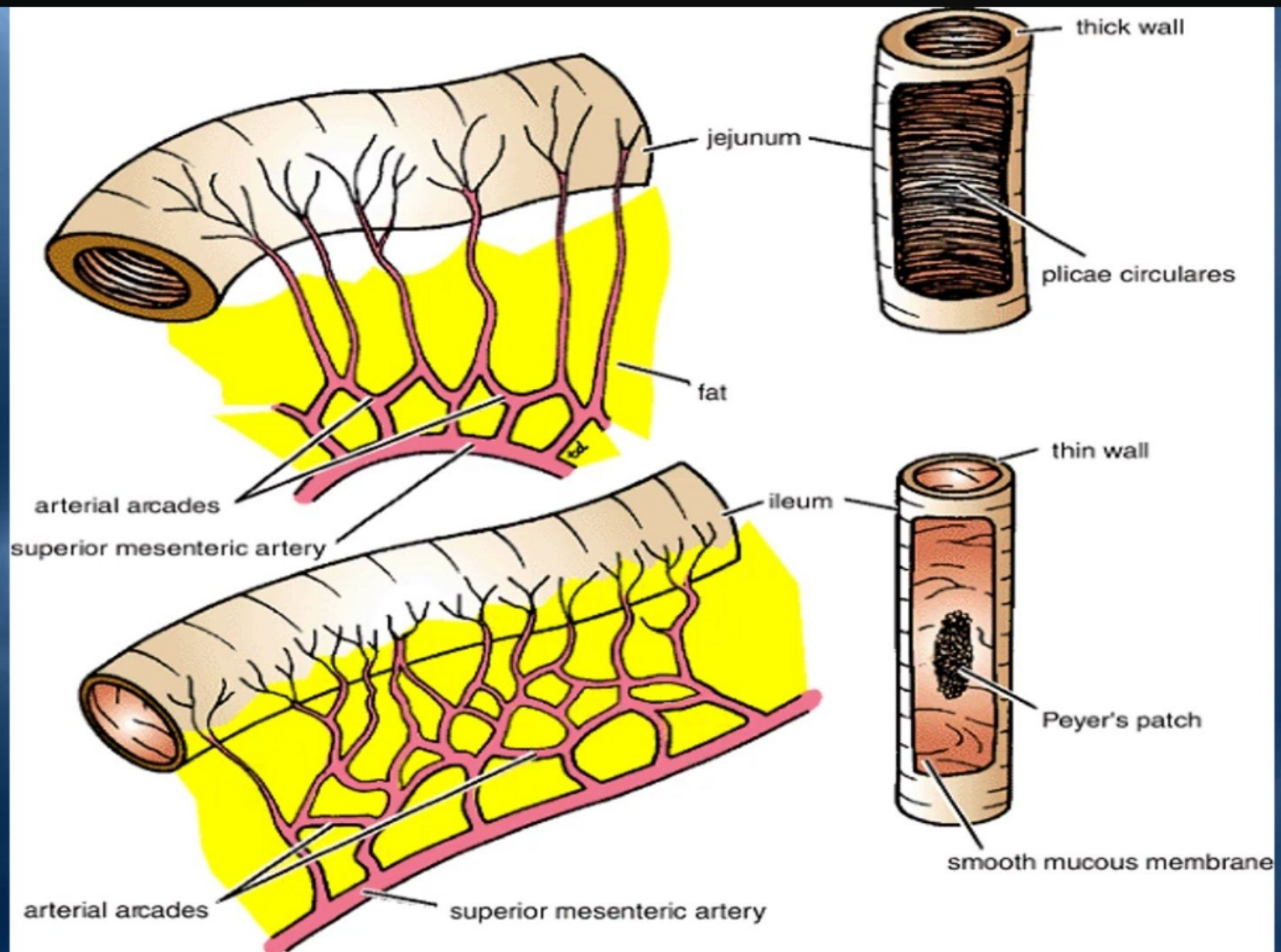


Difference Between Jejunum & Ileum

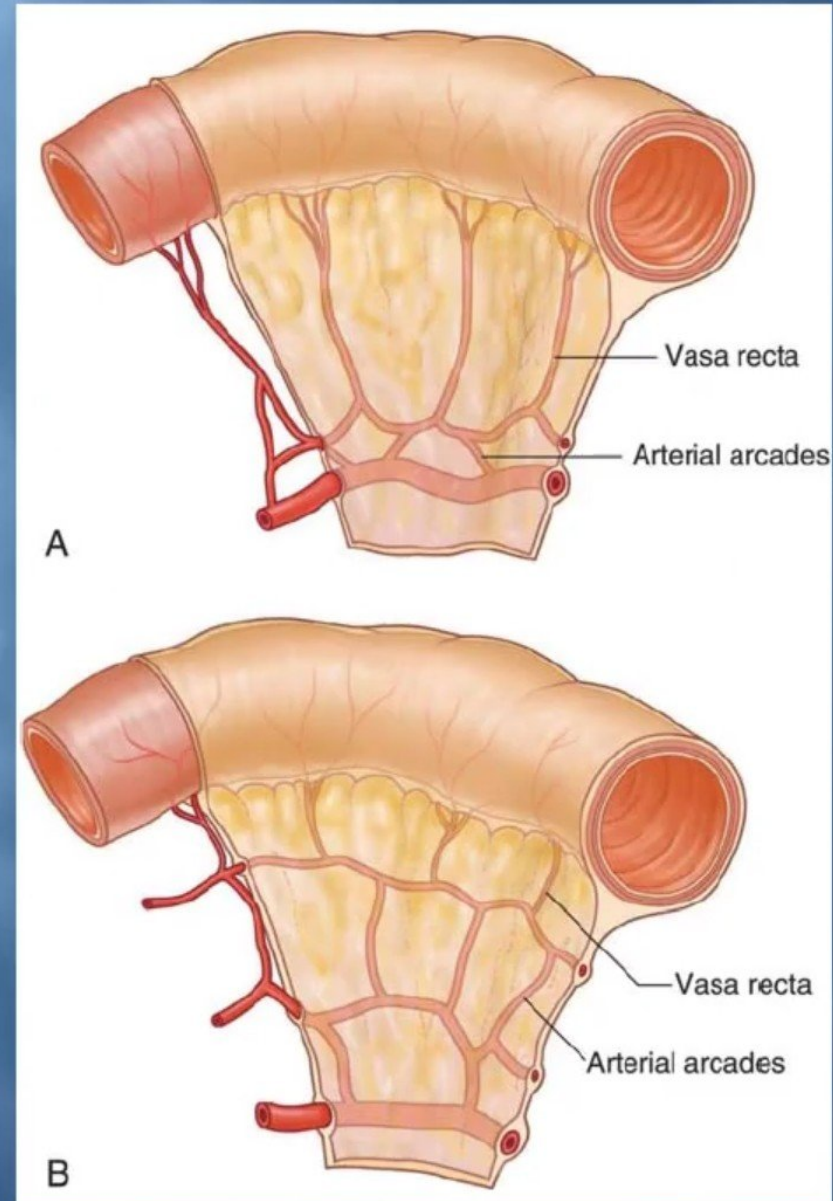
	jejunum	Ileum
length	Proximal 2/5	Distal 3/5
site	in the upper part of the peritoneal cavity below the left side of the transverse mesocolon	in the lower part of the cavity and in the pelvis
wall	thicker wall& redder.	Thinner & less redder.
Arcades in mesentery	- simple ,only one or two arcades - with long infrequent branches - Long vasa recta	- numerous and short terminal vessels - It arise from a series of three or four or even more arcade. - Short vasa recta
Fat in mesentery	- the fat is deposited near the root. - It is scanty near the intestinal wall. - Less in amount → appear window.	- the fat is deposited throughout mesentery - Big amount - No window appear

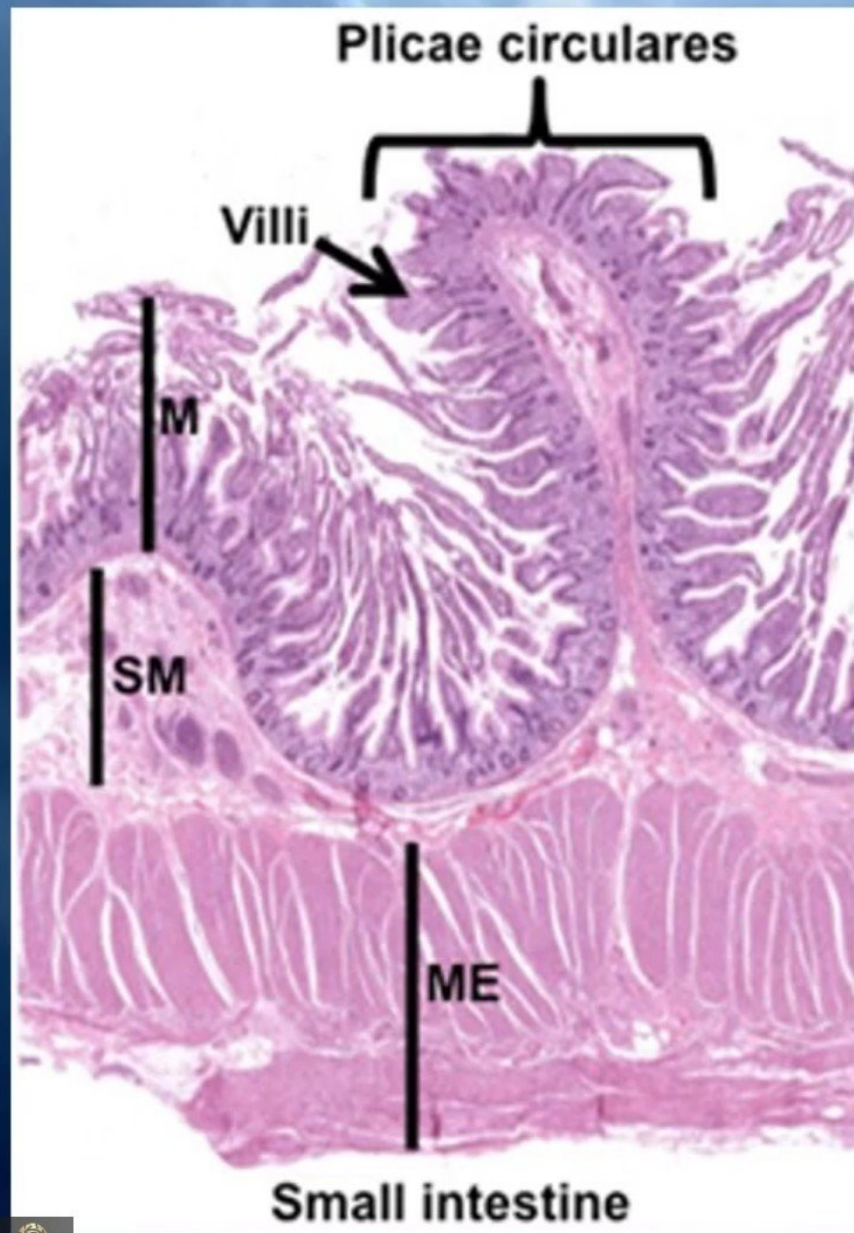
Difference Between Jejunum & Ileum

	jejunum	Ileum
Diameter	wider	smaller
villi	numerous	Less numerous
Plicae circularis (the permanent enfolding of the mucous membrane & submucosa)	They are: 1- larger 2- more numerous 3- closely set	They are: 1- smaller. 2- more widely separated. 3- in the lower part they are absent.
Lymphatic follicles	No or few	Aggregations of lymphoid tissue (Peyer's patches) are present in the mucous membrane

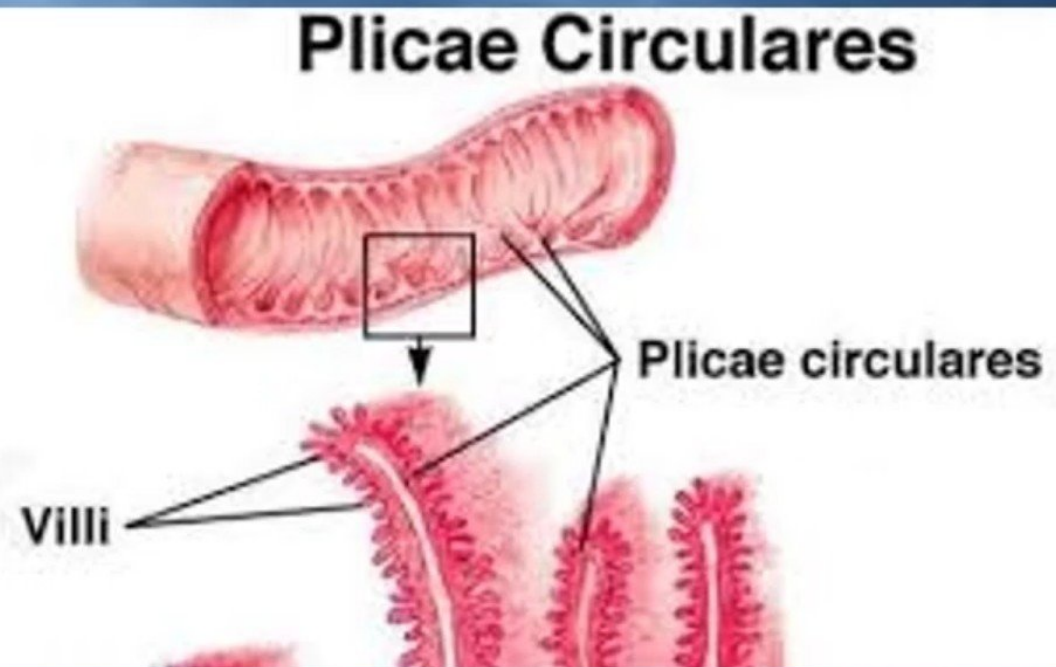


Difference of arterial supply of small intestines





**Plicae Circulares more and large
in Jejunum**



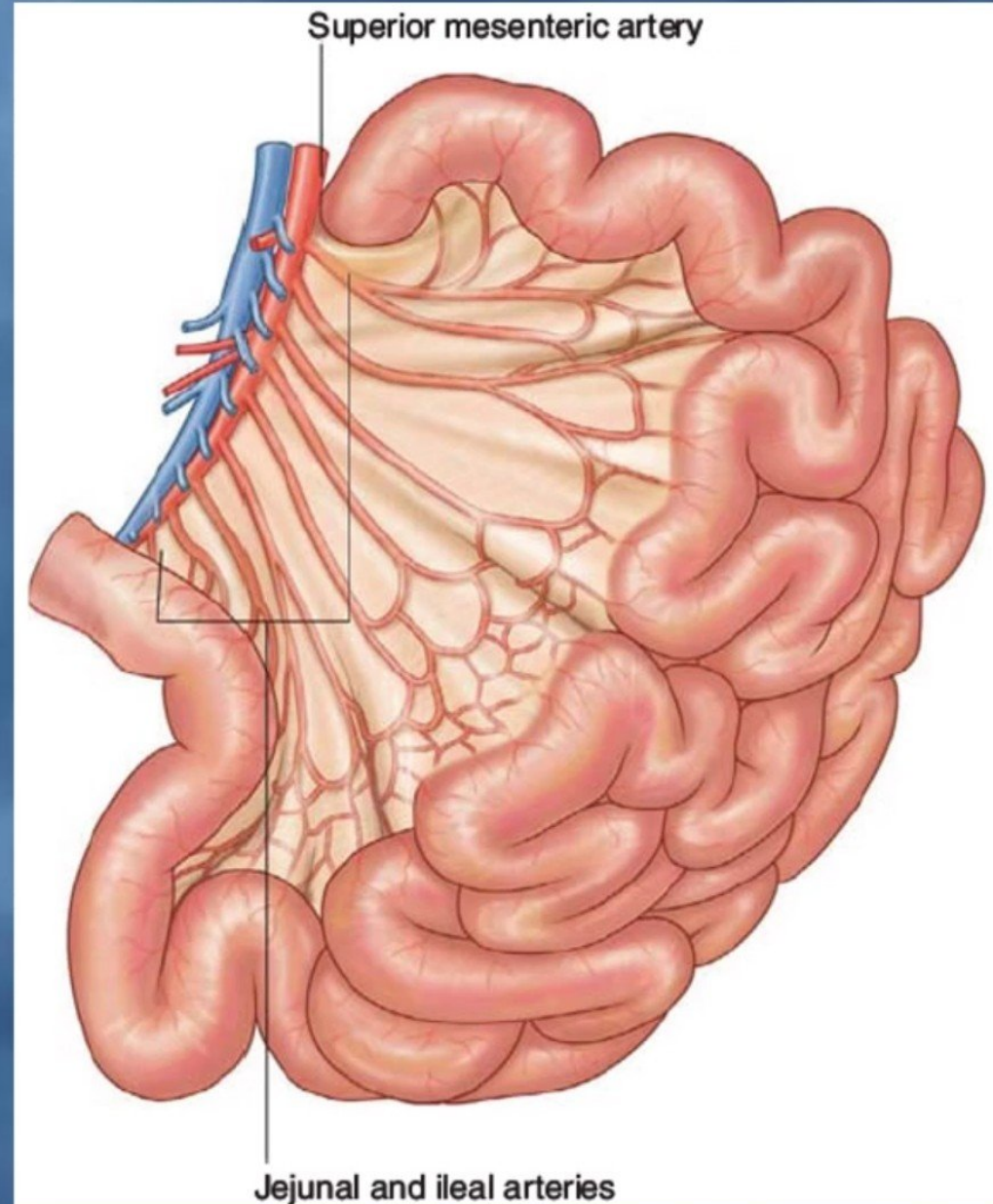
Blood Supply of Jejunum & Ileum

Arteries:

- The arterial supply is from branches of the **superior mesenteric artery**.
- **The intestinal branches** arise from the left side of the artery and run in the mesentery to reach the gut.
- They anastomosis with one another to form a series of **arcades**.
- The lowest part of the ileum is also supplied by **the ileocolic artery**.

Veins:

- **The veins correspond** to the branches of the superior mesenteric artery.
- Drain into **the superior mesenteric vein**.



Nerve Supply of Jejunum & Ileum

- The nerves are derived from **sympathetic** and **parasympathetic (vagus)** nerves from **the celiac and superior mesenteric plexuses**.
- **The parasympathetic vagal** supply augments **peristaltic activity** and **intestinal secretion**.
- There are many afferent fibres whose function is uncertain.
- **The sympathetic** supply, which is **vasoconstrictor** and **normally inhibits peristalsis**, is from **the lateral horn cells of spinal segments T9 and 10**.
- **Pain impulses** use **sympathetic** pathways mainly and
- **Small intestinal pain** is **usually felt** in **the umbilical region of the abdomen**.

Lymph drainage of Jejunum & Ileum

- The lymph vessels follow the arteries and drain upward via **pancreaticoduodenal nodes** to **the gastroduodenal nodes** and then to **the celiac nodes** and downward via **pancreaticoduodenal nodes** to **the superior mesenteric nodes** around the origin of the superior mesenteric artery.

Thanks